

Unit Fractions in Area Models

Answer Key

Grade 2–3 Fractions as Numbers

Quick Answers

1. $\frac{1}{4}$
2. $\frac{3}{3}$
3. $\frac{1}{8}$
4. $\frac{1}{6}$

5. $\frac{1}{5}$
 6. —
 7. $\frac{1}{8}$, $\frac{1}{4}$, $\frac{1}{2}$
 8. —
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Curriculum

Level: Grade 2–3 Fractions as Numbers3.NF.A – *problems 1, 2, 3, 4, 5, 6, 7, 8**Difficulty 1–4 of 5 across 8 tagged checks.*

1. Rectangle cut into four equal parts.

Step 1. Check the parts. Every piece is the same width and the same height, so the cutting is fair and the piece can be named.

Step 2. Count the equal parts across the whole rectangle: one, two, three, four. Four goes on the bottom.

Step 3. One part is shaded, so one goes on top: $\boxed{\frac{1}{4}}$.

Teaching note: say it out loud as “one out of four equal parts.” The words carry the meaning that the symbol only records.

2. Circle cut into three equal parts.

Step 1. The three slices meet at the centre and open by the same amount, so they are equal parts of one whole circle.

Step 2. Count the equal parts: three. Three goes on the bottom.

Step 3. One slice is shaded, so the fraction is $\boxed{\frac{1}{3}}$.

Listen for: “but it is a circle, not a rectangle.” The shape does not matter. What matters is that one whole was cut into equal parts.

3. Square cut into eight equal parts.

Step 1. The square is cut by three up-and-down lines and one across, and all the small rectangles that result are the same size.

Step 2. Count the parts, not the lines: four along the top row and four along the bottom row makes eight equal parts.

Step 3. One part is shaded, so the fraction is $\boxed{\frac{1}{8}}$.

Common wrong answer: $\frac{1}{7}$ — the child counted the cut lines (or counted the parts but skipped one at the end). Have them touch and count each piece.

4. Which bar can be named — equal parts or not.

Step 1. Compare the pieces inside each bar. In bar A every piece is the same length. In bar B the pieces are all different lengths.

Step 2. Only bar A is cut into equal parts, so only bar A's shaded piece has a unit-fraction name. Bar B gets the X.

Step 3. Bar A has six equal parts and one of them is shaded: $\boxed{\frac{1}{6}}$.

Teaching note: bar B still shows a shaded piece of a whole — it is just not *one sixth*, because sixths have to be the same size. This is the idea the whole page rests on.

5. Naming a part that has been moved off the whole.

Step 1. Start with the whole bar, not the strip. The bar is cut into equal parts: one, two, three, four, five. Five goes on the bottom.

Step 2. The strip beside it is exactly the size of one of those parts — lay it against the bar to check.

Step 3. One of five equal parts is $\boxed{\frac{1}{5}}$.

Common wrong answer: $\frac{1}{4}$ — the child counted the parts still sitting in the bar after mentally lifting one out. The bottom number counts how many equal parts the whole was cut into, and that number never changes when a piece is moved.

6. Comparing $\frac{1}{3}$ with $\frac{1}{6}$.

Step 1. The two rectangles are the same size, and each one has exactly one part shaded — so the comparison is only about how big one part is.

Step 2. Cutting a whole into three parts makes bigger parts than cutting the same whole into six. On the page, the top shaded piece is plainly twice as wide as the bottom one.

Step 3. The third is bigger, so $\frac{1}{3} \boxed{>} \frac{1}{6}$.

Listen for: “six is more than three, so the bottom one is bigger.” With unit fractions the bigger bottom number means *more* cuts, and more cuts means *smaller* pieces.

7. Ordering three unit fractions, smallest first.

Step 1. Read each square. Square A is cut into two equal parts, square B into eight, and square C into four. Each has one part shaded, so the three fractions are one half, one eighth and one fourth.

Step 2. All three squares are the same size, so the fraction with the most cuts has the smallest piece. Eight cuts beats four, which beats two.

Step 3. Smallest first, the order is $\boxed{\frac{1}{8}}$, then $\boxed{\frac{1}{4}}$, then $\boxed{\frac{1}{2}}$ — that is B, C, A.

Teaching note: the picture makes this an ordering the child can see before they can reason it. Ask them to point at the smallest shaded piece first, then name it.

8. What Sam got wrong. *Open written explanation — flagged for manual review. The model answer below is what full credit contains.*

Step 1. Look at the three pieces. The shaded one is narrow, the middle one is wide, and the right-hand one is wider still. They are not the same size.

Step 2. A fraction name like one third only works when the whole has been cut into three *equal* pieces. Sam counted the pieces but never checked that they matched.

Step 3. A full-credit sentence says both halves of that: the pieces are not equal, so the shaded piece is not one third — and what would have to change is the cutting. Cut the same bar into three pieces of the same size, and then the shaded piece really would be one third. *Full credit:* the sentence names *equal parts* as the missing condition. “Because it looks wrong” has noticed the problem without naming it — ask the child to finish the thought.